FilterSense

Round 2 prototype progress report | Team Spartans | Thiagarajar College of Engineering, Madurai

Track 2: Water Purifier. Primary: prognostics of filter failure using data analytics. Secondary: inlet water quality sensing beyond TDS.

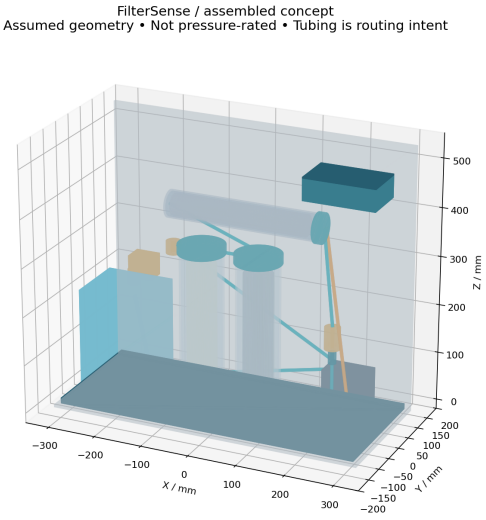
Problem and proposed solution

Fixed time/litre replacement schedules may not reflect variation in water quality and operating conditions. FilterSense combines differential pressure across the pre-filter train, flow decay, turbidity exposure and RO salt rejection to explain deterioration and estimate time to an intervention threshold.

Completed and demonstrated

Version 2 removes forced light backgrounds to preserve text contrast across themes, adds guided reset instructions and an offline interactive 3D viewer. A runnable Streamlit software prototype includes five labelled synthetic scenarios, history replay, CSV export, sensor cards, pressure/flow/TDS plots, causal Kalman smoothing, threshold forecasts and replacement guidance. Leak, invalid-sensor and overpressure fault controls demonstrate latched pump-OFF/valve-CLOSED logic; this is simulation only.

CAD deliverables include 31 STEP part/routing files, an assembled STEP, an exploded STEP, actual geometry renders and dimensioned concept reference sheets. CAD uses assumed dimensions; native SOLIDWORKS import, mating, interference review, fabrication and pressure validation remain pending. The user has confirmed the dashboard opens on their laptop.



Evidence: 25 functional checks and 19 Streamlit interaction checks passed in the development environment. Twelve synthetic held-out runs (60 snapshots) were evaluated. Dashboard trend MAE: 1.21 days; pooled regression baseline MAE: 4.91 days. These are synthetic consistency results, not real-world accuracy.

Costs, sourcing and changes

Reported hardware spending: INR 0 / approximately INR 12,000 planned, based on the supplied project brief. No hardware purchases or laboratory loans are confirmed. Sensor calibration, pressure-rated housings/fittings, pump availability and physical test time are the principal sourcing/execution risks. Proposed next source: existing college laboratory equipment or a supervised borrowed purifier rig.

Change in execution: physical build is deferred because no hardware is available and the team cannot fund the complete setup. This round demonstrates CAD plus software. Revised claim: the prototype computes and explains simulated condition indicators and time-to-threshold estimates; it does not establish actual remaining filter life.

Design review adds the RO reject route and identifies the need for an additional ADC input: the five proposed analogue signals exceed one ADS1115. Estimated feed exposure uses an explicit recovery assumption because product-flow measurement alone is insufficient. P1/P2 diagnose combined restriction, not individual sediment/carbon failure.

Source limitation: the actual submitted Round 1 abstract was not available during preparation. The supplied master brief was used for project details and original cost; team review against the submitted abstract remains pending.

Safety and fail-safe status

Software interlock tests cover leak trip, invalid input, overpressure, latch persistence and reset refusal while fault remains. No physical valve, pump, sensor, pressure relief or electrical protection has been tested. Physical intent: approved pressure-rated wet components, 24 V DC wet zone, protected enclosed supply outside wet zone, leak tray, independent pressure protection and fail-closed valve.

Concept housings are not manufacture-ready pressure vessels. Test water must not be consumed. TDS/turbidity measurements do not certify drinking-water safety. Safety-critical control must remain local and independent of the dashboard.

Round 3 plan

Obtain approved borrowed hardware; verify exact dimensions/interfaces; calibrate sensors and establish clean baselines; verify local safety functions under supervision; collect repeated full degradation runs with measured endpoints; validate on complete held-out physical runs and report errors, false alerts and uncertainty against fixed-time/litre baselines.

Submission status

Prepared deliverables: report, source/CAD package, video script and captioned software/CAD demonstration. No competition upload has been performed. Team should verify playback, confirm consistency with its original abstract and upload one video (maximum 90 s / 50 MB) and this report (2 pages) by 4 November 2026, 11:59 PM IST.